PREDICTION OF STOCK MARKET PRICE USING MACHINE LEARNING

PROJECT REPORT

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BONAFIDE CERTIFICATE

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We affirm that the project work titled **"PREDICTION OF STOCK MARKET PRICE USING MACHINE LEARNING"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Artificial Intelligence and Data Science** is the record of original work done by us under the guidance of **Mrs. Kiruthiga R** , Assistant Professor**,** Department of Artificial intelligence and Data science. It has not formed a part of any other project work(s) submitted for the award of any degree or diploma, either in this or any other University.

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LIST OF ABBREVATIONS

ABBREVATIONS

RNN	RECURRENT NEURAL NETWORK
Zk-SNARKs	ZERO-KNOWLEDGE SUCCINCT NON-INTERACTIVE ARGUMENTS OF KNOWLEDGE
LSTM	LONG SHORT-TERM MEMORY
DR	DIRECT REINFORCEMENT
AI	ARTIFICIAL INTELLIGENCE
GCNN	GRAPH CONVOLUTIONAL NEURAL NETWORK
GED	GRAPH EDIT DISTANCE
LR	LINEAR REGRESSION
SVM	SUPPORT VECTOR MACHINE
POW	PROOF-OF-WORK

STOCK PRICE PREDICTION USING MACHINE LEARNING

ABSTRACT

This study focuses on predicting binary stock price movements—specifically, whether a stock price will move "up" or "down"—using machine learning models, including Linear Regression and Random Forest. The application fetches stock data for major companies, such as Apple, Amazon, Google, Microsoft, and Tesla, using the yfinance library. Leveraging daily stock price data, the models were trained to classify each day based on historical patterns, with the Random Forest model achieving a Mean Squared Error (MSE) of 0.397. This performance highlights the model's relative accuracy in capturing stock movement trends.

A comparison between predicted and actual price labels further showcases the model's classification capabilities, with output probabilities indicating varying confidence levels for each prediction. A candlestick chart visualization was also used to display price trends, complementing the model's predictive insights with a visual representation of historical price movements. The results demonstrate the potential of the Random Forest model to capture complex relationships in stock data, suggesting its viability as a predictive tool for binary stock movement, which could be valuable for investors and market analysts.

KEYWORDS: Stock Prediction, Machine Learning, Random Forest, Linear Regression, Stock Forecasting, Financial Data, Price Movement.

CHAPTER 1

1. INTRODUCTION

Predicting stock price movements has long been a challenging yet attractive endeavor for financial analysts, investors, and researchers. Given the dynamic and often unpredictable nature of the stock market, accurately forecasting whether a stock's price will increase or decrease holds substantial value for making informed investment decisions. Traditional methods for stock prediction have included technical analysis, which focuses on historical price trends and patterns, as well as fundamental analysis, which examines a company's financial health and market position. However, with the rise of machine learning and data analytics, new methods have emerged that leverage vast amounts of historical stock data to detect intricate patterns and trends that may be otherwise imperceptible.

This study explores the application of machine learning models, specifically Linear Regression and Random Forest, to predict binary stock price movements. By transforming the prediction task into a classification problem, the models aim to determine if a stock will go "up" or "down" on a given day based on previous trends. Through the use of these models and evaluation of their outputs, this research seeks to demonstrate the feasibility and potential of machine learning in stock price forecasting, contributing to the growing field of quantitative finance.

1.1 STOCK PRICE PREDICTION

Stock price prediction is a fundamental aspect of financial analytics and investment strategy, where the goal is to accurately forecast the future price of stocks. It relies on analyzing historical price data, market sentiment, trading volumes, and sometimes macroeconomic indicators to understand stock trends and movements. With fluctuations caused by factors such as economic conditions, political events, or company performance, predicting stock prices is challenging and requires a mix of data-driven insights and computational models. Stock price prediction is crucial for investors, portfolio managers, and traders, as precise predictions can maximize returns, manage risk, and inform buying or selling strategies.

1.2 MACHINE LEARNING

Machine learning (ML) is a subset of artificial intelligence focused on building systems that can learn from and make decisions based on data. It is increasingly vital in fields that require data- driven insights, such as stock price prediction, due to its ability to process large amounts of historical and real-time data to detect patterns. In finance, ML techniques such as supervised learning, unsupervised learning, and reinforcement learning allow models to learn relationships in historical stock data to make accurate predictions on future prices. Algorithms like support vector machines, neural networks, and ensemble methods can capture the non-linear and complex dependencies often seen in financial data. By refining these models over time, ML enables predictive models to adapt to market changes and improve with each prediction cycle, ultimately helping investors make informed decisions in an ever-changing financial landscape.

1.3 RANDOM FOREST

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive performance and reduce the likelihood of overfitting. In stock price prediction, Random Forest is advantageous because it can handle high-dimensional data and complex interactions among variables, which are common in financial datasets. This algorithm works by constructing a "forest" of decision trees during training and then averaging their outputs for a final prediction, which makes it more robust and accurate in noisy or complex data environments. In finance, Random Forest can be applied to various tasks such as predicting stock trends, categorizing market behavior, and identifying significant market signals from a multitude of variables. Its ability to perform well with both numerical and categorical data allows it to generate insightful results, making it a popular choice for stock price forecasting.

1.4 LINEAR REGRESSION

Linear regression is a classical statistical method that establishes a linear relationship between a dependent variable and one or more independent variables. It is one of the most widely used models in predictive analytics, particularly in stock price forecasting, where it helps in determining trends based on historical data. Linear regression is often used to predict stock returns, analyze correlations with economic indicators, or forecast general market trends. In finance, it can be employed to examine relationships between stock prices and external factors such as interest rates, inflation, or company earnings. The simplicity and interpretability of linear regression make it a preferred approach when relationships between variables are approximately linear. However, due to its limitations in capturing non-linear relationships, linear regression is often complemented with more complex models in real-world financial forecasting. Nevertheless, it provides a strong foundation and serves as a benchmark for assessing the performance of more sophisticated predictive models.

1.5 FINANCIAL FORECASTING

Financial forecasting involves predicting future economic and financial trends to support decision- making and strategic planning. It encompasses various domains such as revenue projections, economic trend analysis, and stock price prediction. Financial forecasting combines historical data, statistical analysis, and often machine learning techniques to generate predictions that aid in risk management, budget planning, and investment strategies. In stock markets, accurate forecasting helps investors anticipate market fluctuations and identify profitable opportunities. Financial forecasting also plays a key role in organizational budgeting, enabling businesses to allocate resources effectively and plan for potential market shifts. With advancements in data analytics and machine learning, financial forecasting models have become more precise, accommodating large- scale data and complex variable interactions.

1.5 OBJECTIVES

1. Develop a model to accurately predict stock prices using historical data and machine learning algorithms.
2. Compare the effectiveness of Random Forest and Linear Regression in stock price forecasting.
3. Identify the key factors that influence stock price movements and their impact on model accuracy.
4. Enhance model performance by minimizing error rates in stock price prediction.
5. Provide investors with data-driven insights to make informed financial decisions in volatile markets.

CHAPTER 2

2. LITERATURE REVIEW

2.1 CRYPTOCURRENCY LIQUIDITY AND VOLATILITY INTERRELATIONSHIPS DURING THE COVID-19 PANDEMIC

Shane Courbetet.al. Has proposed in this paper, We examine the interactions between cryptocurrency price volatility and liquidity during the outbreak of the COVID-19 pandemic. Evidence suggests that these developing digital products have played a new role as a potential safe-haven during periods of substantial financial market panic. Results suggest that cryptocurrency market liquidity increased significantly after the WHO identification of a worldwide pandemic. Significant and substantial interactions between cryptocurrency price and liquidity effects are identified. These results add further support to the argument that substantial flows of investment entered cryptocurrency markets in search of an investment safe-haven during this exceptional black-swan event.The rapidly developing COVID-19 pandemic generated much confusion with regards to the severity and economic ramifications. The slow and heterogeneous response resulted in substantial variation of virus reproductive rates, manifesting as broad differentials in the magnitude of cross-border cases and subsequent fatalities.

Chinese financial markets have been identified as the initial epicentre of the shock 2020b; Conlon et al., 2020) while international economic contagion effects quickly escalated. Substantial abnormal market pressures, combined with geopolitical pressure and broad concerns based on the severity of the COVID-19 pandemic, contributed to the price of West Texas Intermediate oil falling1 to below -$37 (Courbet et al., 2020a). Investors subsequently struggled to identify credible safe-havens Cryptocurrencies have presented many periods of pronounced price volatility during their rapid growth and development. Since the onset of the COVID-19 pandemic, there is evidence to suggest that cryptocurrencies have played a new role as a potential safe haven during periods of substantial financial market panic. This was buoyed by substantial difficulties in estimating the severity of international issues such as simultaneous COVID-19 and geopolitical pressures. Results suggest that cryptocurrency market liquidity increased, both sharply and significantly, in line with the WHO announcements of a worldwide pandemic.

2.2 CRYPTOCURRENCY TRADING USING MACHINE LEARNING

Thomas E. Kocheret.al. has proposed in this paper We present a model for active trading based on reinforcement machine learning and apply this to five major cryptocurrencies in circulation. In relation to a buy-and-hold approach, we demonstrate how this model yields enhanced risk-adjusted returns and serves to reduce downside risk. These findings hold when accounting for actual transaction costs. We conclude that real-world portfolio management application of the model is viable, yet, performance can vary based on how it is calibrated in test samples.Is active trading viable in cryptocurrency markets and can it yield superior performance relative to a buy-and-hold approach? Using a direct reinforcement (DR) model we demonstrate that, yes, active trading is both viable and profitable in such markets and can yield superior risk-adjusted performance relative to a passive buy-and-hold approach. We show that despite the high volatility risks which cryptocurrency traders face, our DR approach can serve to actively reduce downside risks in their portfolios.The motivation for our DR model stems from two streams of literature.

The first is the growing interest in explaining or forecasting the price behavior of cryptocurrencies and the formation of trading strategies in such markets (Akcora et al. 2018; Bauru and Dimple 2018; Brunei's and Metal 2018; Catania et al. 2019; Phillip et al. 2018). This stream of literature shows how these blockchain-based digital assets have distinct technological and network attributes that set them apart from traditional assets. In addition, this stream of literature also demonstrates the growing need in understanding whether active trading in such markets is viable. For example, Tzouvanas et al. (2020) show momentum trading may be profitable but only in the short-run, while Grubs and Sopot (2019) provide evidence that momentum trading is not profitable when examining a large cross-section of cryptocurrencies.We show how reinforcement machine learning

can make cryptocurrency trading decisions that optimize actively managed portfolios. Our results compare favorably to a naive buy-and-hold approach. Theoretically, these results provide some preliminary evidence that cryptocurrency prices may not follow a purely random walk process. This is because we show how our active trading strategy can yield superior risk-adjusted returns for investors and even reduce portfolio downside risk across time and if the model is trained adequately. Thus, the possibility of devising novel active trading strategies in the future, based on some quantitative or theoretical framework, may be useful in cryptocurrency markets instead of relying on naive buy-and-hold approaches.

2.3 BLOCKCHAIN FOR SECURING AI APPLICATIONS AND OPEN INNOVATIONS

Rocha Shinedet.al. Has proposed in this system, Nowadays, open innovations such as intelligent automation and digitalization are being adopted by every industry with the help of powerful technology such as Artificial Intelligence (AI). This evolution drives systematic running processes, involves less overhead of managerial activities and increased production rate. However, it also gave birth to different kinds of attacks and security issues at the data storage level and process level. The real-life implementation of such AI-enabled intelligent systems is currently plagued by the lack of security and trust levels in system predictions. Blockchain is a prevailing technology that can help to alleviate the security risks of AI applications. These two technologies are complementing each other as Blockchain can mitigate vulnerabilities in AI, and AI can improve the performance of Blockchain. Many studies are currently being conducted on the applicability of Blockchain for securing intelligent applications in various crucial domains such as healthcare, finance, energy, government, and defense.

However, this domain lacks a systematic study that can offer an overarching view of research activities currently going on in applying Blockchain for securing AI-based systems and improving their robustness. This paper presents a bibliometric and literature analysis of how Blockchain provides a security blanket to AI-based systems. Two well- known research databases (Scopus and Web of Science) have been examined for this analytical study and review. The research uncovered that idea proposals in conferences and some articles published in journals make a major contribution. However, there is still a lot of research work to be done to implement real and stable Blockchain-based AI systems.In the present era, the vast use of electronic gadgets, social media, and automation causes a voluminous amount of data to be produced per second. At the same time, there is a rise in cyber-attacks such as identity thefts, data

breaches, etc. There are various cybersecurity measures that deal with such kinds of attacks. Blockchain technology is an emerging trend in the world of the internet and digitalization, providing high level security. The available security measures are based on centralized servers/systems. Here, ingle points of failure, susceptibility to security breaches, and the need for trusted third parties are all disadvantages. Instead, Blockchain technology is a decentralized system wherein the trusted third parties are absent, and trust is established within the nodes available in the network. Initially, Blockchain came into the picture in the form of cryptocurrencies such as Bitcoin, Ether, and Ripple.

After understanding the potential features of Blockchain, many researchers came with the idea of applying Blockchain technology to various industrial applications such as voting, healthcare, banking, and supply chain management, to meet integrity, availability and confidentiality requirements, without any involvement of a central authority.This study highlighted that Blockchain technology is not just for financial transactions or cryptocurrencies. Rather, Blockchain is seen as an emerging technology for securing crucial applications. Since 2019, there has been massive growth in the publications in the Blockchain domain. Many countries contribute to the research of Blockchain and its applicability, but China and the United States are leading in the contribution.

2.4 CODE CHARACTERIZATION WITH GRAPH CONVOLUTIONS AND CAPSULE NETWORKS

Purnima Harridanset.al. Has proposed in this system We propose SiCaGCN, a learning system to predict the similarity of a given software code to a set of codes that are permitted to run on a computational resource, such as a supercomputer or a cloud server. This code characterization allows us to detect abusive codes. Our system relies on a structural analysis of the control-flow graph of the software codes and two different graph similarity measures: Graph Edit Distance (GED) and a singular values based metric. SiCaGCN combines elements of Graph Convolutional Neural Networks (GCN), Capsule networks, attention mechanism, and neural tensor networks. Our experimental results include a study of the trade-offs between the two similarity metrics and two variations of our learning networks, with and without the use of capsules. Our main findings are that the use of capsules reduces mean square error significantly for both similarity metrics. Use of capsules reduces the runtime to calculate the GED while increases the runtime of singular values calculation the era of exascale computing code characterization is extremely important for super-

computing centers and cloud vendors. Especially, for the institutions such as the US DOE national labs that run scientific codes (such as hydrodynamics simulations, molecular dynamics, etc). In some cases, these codes appear similar to the allowed scientific codes while abusive algorithms (for example, Bitcoin mining) run underneath, misusing the compute resources. In fact, one of the illicit crypto-currency mining is the use of computers connected to internet in order to run the mining codes. The first analysis appeared in 2014 that analyzed thousands of public transactions in order to detect the bitcoin mining. Since 2014, these Recently, these threats have increased significantly, a detailed survey of crypto-mining is in. To highlight the seriousness of the problem, recently in May 2020, cryptocurrency miners have attacked the highly secure European supercomputers. Moreover, we further want to employ unsupervised methods such as the community detection strategies in order to find the clusters of programs and separate the programs that are abusive. Another interesting unsupervised direction will be to incorporate large scale nonnegative low rank representations of programs graphs in order to detect the outliers, application of heuristics in order to automatically generate the deep network architectures for abusive program detection.

2.5 TAMING THE BLOCKCHAIN BEAST REGULATORY IMPLICATIONS FOR THE CRYPTOCURRENCY MARKET

Savva Shanaeet.al. Has proposed in this system, This paper uses a unique dataset of 120 regulatory events from five classes to test the relevance of the regulatory framework for cryptocurrency value. Time-series market-wide estimates and panel estimates for 300 individual coins and tokens show statistically and economically significant impact of anti-money laundering and issuance regulation. Tighter regulation and more active role of government decrease cryptocurrency prices, evidencing that potentially lower risks and wider adoption commonly attributed to the establishment of the regulatory framework do not compensate for respective efficiency and consumer utility losses. The market is generally efficient in reflecting regulatory information in cryptocurrency prices.With the advent of blockchain technology in general and its most famous practical applications, cryptocurrencies, in particular, the issue of its regulation has been becoming increasingly relevant. Academics, policymakers, investors and blockchain enthusiasts alike engage in the heated debate on what is the socially desirable level of cryptocurrency regulation. Some of the most vocal opponents, notably Nobel laureate Joseph Stieglitz, call for a complete

cryptocurrency ban (Costello, 2017), while others advocate for a more or less flexible regulatory framework (Greenberg, 2011; Bullen, 2013; Atmore, 2015). On the other side of the opinion spectrum, it is argued that cryptocurrencies should not be regulated at all (Davidson and Block, 2015; Swan, 2015).

The overall relevance of cryptocurrency regulation is overwhelmingly supported by the fact that even the Bank for International Settlements periodically issues thematic articles on the topic.The heterogeneity of views on the issue and lack of consensus can be easily explained as cryptocurrencies present both significant technological advantages and sources of consumer value as well as notable and non-trivial challenges and concerns. As such, the potential consumer value of cryptocurrencies has been widely studied in terms of alternative money supply or payment service provision.This study has estimated the impact of five broad regulatory event classes on the cryptocurrency market, using manually collected data on 120 events and a sample of 300 coins and tokens on a daily basis. Having applied various event studies techniques in the time-series and panel data models, it has shown that news associated with the tightening and, generally, further development of anti-money laundering, exchange and issuance regulation as well as news on the successful launch of state-backed sovereign cryptocurrencies have a statistically and economically significant negative effect on the market. Further, the relaxation of the respective policy measures and the declaration of hands-off, self-regulatory approach as well as rejections or heavy critiques of state-backed blockchain payment system projects, contrastingly, lead to coin prices appreciating. The results are extremely consistent for anti-money laundering and issuance regulation policies and rather consistent for exchange regulation and state-backed issuance.

2.6 CRYINGJACKPOT: NETWORK FLOWS AND PERFORMANCE COUNTERS AGAINST CRYPTOJACKING

Gilberto Gomeset.al has proposed in this paperCrypto jacking, the appropriation of users' computational resources without their knowledge or consent to obtain cryptocurrencies, is a widespread attack, relatively easy to implement and hard to detect. Either browser-based or binary, crypto jacking lacks robust and reliable detection solutions. This paper presents a hybrid approach to detect crypto jacking where no previous knowledge about the attacks or training data is needed. Our Crypto jacking Intrusion Detection Approach, CRYINGJACKPOT, extracts and combines flow and performance counter-based features, aggregating hosts with similar behavior by using

unsupervised machine learning algorithms. We evaluate CryingJackpot experimentally with both an artificial and a hybrid dataset, achieving F1-scores up to 97%.Bitcoin's presentation in 2009 made headlines, being the starting point of what would be the popularization of crypto-mining and with this the dissemination of a new attack, crypto jacking.

This recent threat is defined as the illicit appropriation of users' computational power without their knowledge or consent to mine cryptocurrencies. This attack can be performed through two major attack vectors: browser-based scripts and binary malware (executables). Bitcoin created a new paradigm: an alternative to traditional electronic payment systems based on intermediaries (e.g., banks). Relying on such intermediaries is susceptible to problems like potentially misplacing trust, additional costs (money transfer fees), and operation delays. Bitcoin, a digital representation of assets, alongside with blockchain, an append-only universally accessible ledger of transactions, came to replace those transaction intermediaries using a decentralized network of nodes. For a new group of transactions, intensive computing power must be used to validate, add, and distribute a new block with those transactions through the blockchain network. The solution to a cryptographic problem must be found by computing a so-called Proof of Work (PoW). The miner that obtains the PoW and manages to append the block to the blockchain is rewarded with new coins.We present Cryingjackpot, a hybrid approach for crypto jacking detection, based on unsupervised learning, that detects undefined attacks without signatures and clean training data. CRYINGJACKPOT is based on clustering, i.e., on aggregating hosts with similar traffic patterns, using for this end the combination of features from flow-based data and performance indicators for Windows OS machines.

2.7 BITCOIN PRICE PREDICTION USING ENSEMBLES OF NEURAL NETWORKS

Edwin Sin et.al has proposed in this paper This paper explores the relationship between the features of Bitcoin and the next day change in the price of Bitcoin using an Artificial Neural Network ensemble approach called Genetic Algorithm based Selective Neural Network Ensemble, constructed using Multi-Layered Perceptron as the base model for each of the neural network in the ensemble. To better understand the practicality and its effectiveness in realworld application, the ensemble was used to predict the next day direction of the price of Bitcoin given a set of approximately 200 features of the cryptocurrency over a span of 2 years. Over a span of 50 days, a trading strategy based on the ensemble was compared against a "previous day trend following"

trading strategy through back-testing. The former trading strategy generated almost 85% returns, outperforming the "previous day trend following" trading strategy which produced an approximate 38% returns and a trading strategy that follows the single, best MLP model in the ensemble that generated approximately 53% in returns.The digital cryptocurrency, Bitcoin, runs on an online decentralized network with no dependency on any government or legal entity as it relies heavily on peer-to-peer networking and cryptography to maintain its integrity.

This "trust-less" system facilitates ease of transactions of goods and services with low transaction costs associated. As the market is relatively new, existing works related to forecasting in this market is fairly limited. One study showed that Google Trends data and volume of tweets related to Bitcoin on Twitter have positive correlation with Bitcoin's price and hence may be able to predict the fluctuations in price of Bitcoin. In another study, Bayesian Regression, a binary classification algorithm, was used to predict price variation in Bitcoin and the prediction gave almost 200% returns in less than 60 days when used with a trading strategy. The study concluded that there may be 'information' in Bitcoin's historical data that can help predict future price variations.

2.8 PREDICTION OF THE PRICE OF ETHEREUM BLOCKCHAIN CRYPTOCURRENCY IN AN INDUSTRIAL FINANCE SYSTEM

Poongodi M et.al has proposed in this paper Cryptocurrency has gained considerable popularity in the past decade. The untraceable and uncontrolled nature of cryptocurrency attracts millions of people around the world. Research in cryptocurrency is dedicated to finding the ether and predicting its price according to the cryptocurrency's past price inflations. In this study, price prediction is performed with two machine-learning methods, namely linear regression (LR) and support vector machine (SVM), by using a time series consisting of daily ether cryptocurrency closing prices. Different window lengths are used in ether cryptocurrency price prediction by using filters with different weight coefficients. In the training phase, a cross-validation method is used to construct a high-performance model independent of the data set. The proposed model is implemented using two machine learning techniques. When using the proposed model, the SVM method has a higher accuracy (96.06%) than the LR method (85.46%). Furthermore, the accuracy score of the proposed model can be increased up to 99% by adding features to the SVM method.Over the past decades, the world has seen multiple technological advancements, such as the Internet of things (IoT), ecommerce, and digital payments. These advancements not only affect

how one interacts with the world but also change how one interacts and exchanges money for requested services. Nowadays, banking, transaction, and other e-commerce activities are becoming more advanced due to new developments in technology. In the past 20 years, due to the trend of moving from printed currency to virtual currency, a new type of currency called cryptocurrency has emerged . Cryptocurrency is basically a digital currency that uses advanced encryption techniques for regulating its currency units.

Cryptographyhas been applied to make payments more secure and to verify transactions. Verification of digital currency depends on the transaction process. Cryptocurrency is used to transfer funds electronically without any interference of a central entity, such as a bank . The flow of such transactions is controlled with the help of cryptocurrency.Price prediction can provide stock brokers and cryptocurrency traders with an upper hand in the market. It can also help miners mine coins for maximum profit. The algorithms used yielded highly accurate results. Therefore, the trained model can be deployed and implemented in real time. Compared with the SVM algorithm, LR exhibits a higher accuracy on the window, especially for a large dataset. The LR algorithm works so well because the last three mean values come with sufficient information for the prediction of the next mean value.

2.9 TOWARDCHARACTERIZING BLOCKCHAIN-BASED CRYPTOCURRENCIES FOR HIGHLY-ACCURATE PREDICTIONS

Muhammad Saad et.al has proposed in this system Recently, the Blockchain-based cryptocurrency market witnessed enormous growth. Bitcoin, the leading cryptocurrency, reached all-time highs many times over the year leading to speculations to explain the trend in its growth. In this paper, we study Bitcoin and Ethereal and explore features in their network that explain their price hikes. We gather data and analyze user and network activity that highly impact the price of these cryptocurrencies. We monitor the change in the activities over time and relate them to economic theories. We identify key network features that help us reason about the determine the demand and supply dynamics in a cryptocurrency. Finally, we use machine learning methods to construct models that predict Bitcoin price. Based on our experimental results using two large datasets for validation, we confirm that our approach provides an accuracy of up to 99% for Bitcoin and Ethereal price prediction in both instances.Blockchain-based digital currencies have witnessed enormous change in value over the last few years.

years. However, today, the cryptocurrency world has more than 5000 cryptocurrencies and more than 5.8 million active users. In this paper, we carry out a study on Bitcoin and Ethereal to analyze their network features that capture the user behavior and in turn have an impact on their price. The underlying technology of every cryptocurrency is the Blockchain.

Blockchain acts as a decentralized public database that preserves anonymity and augments trust between the users. Trust in an anonymous peer-to-peer model is achieved by consensus protocols such as Proof-of- Work (PoW), Proof-of-Stake (PoS), Proof-of-Knowledge (PoK), and distributed consensus. The decentralized environment and the append-only model prevent Blockchains failure and datatampering, and such features lay ideal foundations for cryptocurrency applications to be built on top of Blockchain.In this paper, look into analyzing cryptocurrency market price through a correlation analysis with various cryptocurrency attributes, exemplified by Bitcoin and Ethereum. We collect data spanning more than 20 months and estimate the most significant features that influence the price. We computed the correlation between features such as hash rate, number of users, transaction rate, total bitcoins and price. We map the change in features on users and network activities to understand the dynamics of the cryptocurrencies. We used our findings to construct a machine learning model that accurately predicts Bitcoin and Ethereum prices with the minimum error.

2.10 A NOVEL APPROACH FOR ABO BLOOD GROUP PREDICTION USING FINGERPRINT THROUGH OPTIMIZED CONVOLUTIONAL NEURAL NETWORK

Vijay Kumar Patil et.al has proposed in this paper The fingerprints of humans hold many potentials and having plenty of unique characteristics. It is one of the primary diagnostic tools used from accent's era because of their distinctive identity. It opens a lot of possibilities for human science research, by analyzing fingerprints, some researchers have tried to predict an individual's gender or age. In the 20th century, fingerprint identification and analysis have become commonplace also it has evolved into a crucial component of forensic at crime scenes. Similarly, like a fingerprint, the blood group is also unique for each individual. This study focuses on ABO and Rh systems, which are among the most prominent blood grouping methods. This paper proposed an optimized Convolutional Neural Network (CNN) which is designed as an extension of an Alex Net, that correlates the fingerprint patterns or different features of the fingerprint with the blood group of an individual. Researchers have only attempted to connect fingerprint patterns with blood types

prior to this proposed method. The result and performance of proposed CNN framework is compared with three different CNN variations like LeNet-5, ZFNet, and Alex Net. The design of proposed CNN used for the prediction of the blood group having noticeable performance with 95.27 % accuracy rate.In 1926, Dr. Harold Cummins introduced a study of fingerprint patterns, but it is even now it is used earlier than several hundred years ago. For almost a century, fingerprints have been utilized as a form of personal identification. Because of its uniqueness and persistence, it is currently widely used in biometric authentication. Although fingerprints may be classified into four fundamental categories: arch, tented arch, loop which is categorized as left loop and right loop and whorl, shown in Fig. 1. Furthermore, fingerprint made of ridges and valleys like minutia patterns. Ridge ends and ridge bifurcations are the two main minutia characteristics considered in fingerprint recognition.

Other characteristics are used as well. Automatic fingerprint identification systems are divided into three categories based on the characteristics involved in fingerprint recognition: minutiae-based, image-based, and ridge feature-based approaches. The Galton's details consist of more than 100 interleaved ridge and valley patterns on a single rolled fingerprint. For various identification purposes, fingerprints can be used due to their changeability during a person's lifetime and their uniqueness.Fingerprints are an effective way of identifying people because of their immense potential. A novel minutiae-based approach is proposed to identify blood groups. Additionally, fingertip patterns of individuals have multiple minutiae, it is very much feasible to exact maximum minutia from fingerprints if it must be taken with a digital equipment. Blood group identification is becoming a more appealing option because to recent advancements in fingerprint detecting and image processing technology and advances in the accuracy and speed of recent algorithms to identifying approaches that have been used having accuracy with 95.27 %.

CHAPTER 3

SYSTEM ANALYSIS

Blockchain technology is becoming increasingly popular because of its applications in various fields. It gives an edge over the traditional centralized methods as it provides decentralization, immutability, integrity, and anonymity. The most popular application of this technology is crypto currencies, which showed a massive rise in their popularity and market capitalization in recent years. Individual investors, big institutions, and corporate firms are investing heavily in it. However, the crypto market is less stable than traditional commodity markets. It can be affected by many technical, sentimental, and legal factors, so it is highly volatile, uncertain, and unpredictable. Plenty of research has been done on various crypto currencies to forecast accurate prices, but the majority of these approaches cannot be applied in real-time. Motivated from the aforementioned discussion, in this paper, we propose a deep-learning-based hybrid model (includes Gated Recurrent Units (GRU) and Long Short Term Memory (LSTM)) to predict the price of Lite coin and Zcash with inter-dependency of the parent coin. The proposed model can be used in real-time scenarios and it is well trained and evaluated using standard data sets. Results illustrate that the proposed model forecasts the prices with high accuracy compared to existing models.

3.1.1 DRAWBACKS

- Deep learning models require large amounts of data to train effectively. This data can be difficult to obtain and may not be representative of the real market.
- It can be computationally expensive to train and deploy. This can make them impractical for use by individual investors and traders.
- This is difficult to interpret, making it difficult to understand why they make the predictions that they do. This can make it difficult to trust their predictions, especially when making large investment decisions.
- It can over fit the training data, which means that they learn the training data too well and are unable to generalize to new data. This can lead to inaccurate predictions on new data.

3.2 PROPOSED SYSTEM

The proposed system aims to enhance stock price prediction by integrating advanced machine learning algorithms, specifically Random Forest and Linear Regression, to analyze historical market data and identify key patterns influencing price movements. The system preprocesses the data by cleaning, normalizing, and selecting important features while mitigating overfitting, ensuring robust and accurate predictions. Random Forest, through ensemble learning, captures complex data patterns, while Linear Regression provides insights into linear relationships between historical prices and variables. Additionally, the system includes a visualization module that generates candlestick charts and comparison graphs to help users interpret predictions alongside actual market trends. By focusing on minimizing prediction errors and optimizing performance, the system equips investors with data-driven insights for making informed, strategic decisions in the stock market, ultimately offering an intelligent tool for reliable forecasting and enhanced decision-making.

3.2.1 ADVANTAGES

- Combining Random Forest and Linear Regression ensures precise stock price predictions by capturing both complex and linear data patterns.
- Real-time insights from the system help investors make data-driven, informed decisions, reducing the risks of manual analysis.
- The system is adaptable to different financial markets and can be expanded to include additional features for broader market analysis.
- Candlestick charts and prediction comparisons provide clear, easy-to-understand graphical insights for better market analysis.
- The model's preprocessing techniques improve the speed and accuracy of predictions while handling large datasets effectively.

3.3 FEASIBILITY STUDY

The feasibility study for the proposed stock market price prediction system evaluates its technical, operational, and economic viability. Technically, the system leverages advanced machine learning algorithms, such as Random Forest and Linear Regression, to accurately predict stock prices based on historical data, ensuring compatibility with existing infrastructure. Operationally, it can be seamlessly integrated into financial institutions or trading platforms, providing users with automated and real-time predictions to aid in decision-making. Economically, the system promises a high return on investment by reducing reliance on manual analysis, improving market prediction accuracy, and streamlining the decision-making process. The feasibility study demonstrates that the proposed system is both achievable and beneficial, given the available resources and technology.

- Technical Feasibility
- Operation Feasibility
- Economic Feasibility

3.3.1 TECHNICAL FEASIBILITY

The technical feasibility of the proposed stock market price prediction system is centered on utilizing machine learning algorithms, such as Random Forest and Linear Regression, which are well-suited for predicting market trends based on historical data. These algorithms are backed by robust mathematical models and are compatible with existing data analytics platforms, making them viable for integration into real-time trading systems. The system will require access to large datasets, which can be sourced from financial market APIs or databases, and can be processed using scalable cloud computing infrastructure. Additionally, the system's ability to handle continuous data streams and perform real-time predictions ensures that it can meet the dynamic and fast-paced demands of the stock market. The use of these proven technologies assures that the system is technically feasible and can be implemented efficiently with the available resources.

3.3.2 OPERATIONAL FEASIBILITY

The operational feasibility of the stock market price prediction system is based on its ability to integrate smoothly into the existing operations of financial institutions, investment

users to interpret the predicted stock prices and make informed decisions. It leverages well-established machine learning algorithms, ensuring that the prediction process is automated and efficient. Furthermore, the system can be continuously updated with new market data, providing real-time insights. Its implementation will streamline the decision-making process by offering actionable predictions, which can be integrated into existing trading strategies. With proper infrastructure support, the system can be seamlessly incorporated into daily trading operations without disrupting the current workflow, ensuring high operational efficiency and effectiveness.

3.3.3 ECONOMIC FEASIBILITY

The economic feasibility of the stock market price prediction system hinges on its potential to provide a significant return on investment (ROI) by improving the accuracy of trading decisions and optimizing market strategies. The system's cost is primarily associated with data acquisition, computational resources, and system maintenance, which are manageable given the increasing availability of financial data and affordable cloud computing options. Additionally, by automating the prediction process and reducing human error, the system can increase the speed and efficiency of decision-making, potentially leading to higher profits and reduced financial risks. Over time, the economic benefits, including improved market predictions and cost savings from operational efficiencies, are expected to outweigh the initial setup and operational costs, making the system a financially viable solution for users in the financial market.

CHAPTER 4

SYSTEM SPECIFICATION

4.1 HARDWARE REQUIREMENTS

- Processor Type : AMD RYZEN 7
- Speed : 4.40GHZ
- RAM :16 GB RAM
- Hard disk : 1 TB
- Keyboard : 101/102 Standard Keys
- Mouse : Optical Mouse

4.2 SOFTWARE REQUIREMENTS

- Operating System : Windows 10
- Front End : Flask
- Coding Language : Python

CHAPTER 5

SOFTWARE DESCRIPTION

FRONT END: PYTHON IDLE

Python IDLE (Integrated Development and Learning Environment) serves as a simple yet effective integrated development environment (IDE) that comes bundled with Python installations. It's designed to offer a minimalistic but functional interface for writing Python code, making it especially suitable for beginners or those who prefer a straightforward development setup. Despite its simplicity, IDLE has several features that support the development process, including but not limited to:

Features of Python IDLE

- Interactive Interpreter: Allows direct execution of Python commands in a REPL (Read-Eval-Print Loop) environment. This feature is particularly useful for learning Python syntax and for quick tests and calculations.
- Syntax Highlighting: Highlights different elements of Python syntax (such as keywords, variables, strings) in different colors, improving code readability and making it easier to identify syntax errors or typos.
- Auto-completion: Offers suggestions for variable names, functions, and methods as you type, based on the Python language and the code you've already written. This feature speeds up coding and reduces typing errors.
- Integrated Debugger: Comes with a simple debugger that allows you to execute your code step by step, inspect variables, and identify the source of errors. This tool is crucial for learning how to debug programs effectively.
- Simple Editor: IDLE's editor provides basic functionality for writing code, including indentation, line numbering, and a search/replace feature. While it might lack advanced features found in more sophisticated IDEs, it's more than capable for beginners and intermediate-level projects.

- Customization: Allows you to customize aspects of the IDLE environment to suit your preferences, including the font size, type, and the color scheme used for syntax highlighting.

Target Audience and Use Cases

- Beginners: IDLE is particularly well-suited for beginners because of its simplicity and the immediate feedback it offers through the interactive interpreter. It's an excellent tool for learning Python syntax and concepts.
- Education: In educational settings, IDLE is a popular choice for teaching Python, as it doesn't overwhelm students with complex features and focuses on the basics.
- Small to Medium Projects: For small to medium-sized projects or for scripting purposes, IDLE provides a straightforward and efficient environment. It's also useful for quick tasks, experiments, or for those who prefer a minimal setup.

Comparing IDLE with Other IDEs

While IDLE is suitable for a range of purposes, some developers might find it lacking for more complex projects. Other IDEs like PyCharm, Visual Studio Code, or Atom offer more advanced features such as integrated version control, a wider array of plugins, advanced debugging tools, and better support for web development, data analysis, and application development.

CHAPTER 6

PROJECT DESCRIPTION

6.1 PROBLEM DEFINITION

The problem addressed by the stock market price prediction system is the inherent uncertainty and volatility in financial markets, which makes it challenging for investors and traders to make accurate predictions and informed decisions. Traditional methods of forecasting stock prices often rely on historical data and basic statistical models, which can fail to capture the complex patterns and factors influencing market movements. This can lead to poor investment strategies, financial losses, and missed opportunities. The goal of the proposed system is to leverage advanced machine learning techniques, such as Random Forest and Linear Regression, to improve the accuracy of stock price predictions, providing more reliable insights for decision-making in dynamic and unpredictable financial environments.

6.2 MODULE DESCRIPTION

1. DATA COLLECTION AND INTEGRATION MODULE

This module is responsible for fetching historical stock data for major companies like Apple, Amazon, Google, Microsoft, and Tesla. The yfinance library is used to collect stock data, including daily stock prices (open, close, high, low) and volume. The data is integrated into a unified dataset, which serves as the foundation for further analysis and prediction tasks. The module ensures that the dataset is continuously updated for accurate forecasting.

2. DATA PREPROCESSING AND CLEANING MODULE

Data preprocessing is a critical step in preparing the collected stock data for machine learning. This module handles tasks like cleaning the data by removing any missing or null values, handling outliers, and ensuring data consistency. It also normalizes the data to bring all features into a comparable range and formats the data into time-series format suitable for training the models. The cleaned data is then split into training and testing datasets to evaluate model performance effectively.

3. FEATURE ENGINEERING AND SELECTION MODULE

In this module, new features are generated from the raw stock data to help improve the model's prediction capability. The features could include technical indicators like moving averages, volatility measures, and price momentum indicators. Feature selection techniques are used to identify the most relevant features by analyzing correlations and eliminating redundant or less useful variables. This module ensures that only the most informative features are used for training, optimizing the models for better accuracy.

4. MODEL TRAINING AND VALIDATION MODULE

The heart of the system, this module is responsible for training the two core algorithms: Linear Regression and Random Forest. The system uses Linear Regression to model the relationship between historical stock prices and the target variable, predicting future stock prices based on linear trends. The Random Forest algorithm, linear regression, is used to capture more complex patterns in the data by building multiple decision trees and aggregating their results. The module trains both models on the pre-processed data, tunes their parameters, and validates them using performance metrics like Mean Squared Error (MSE) and R-squared.

5. PREDICTION AND OUTPUT GENERATION MODULE

After training, the models are used to generate predictions for future stock prices. The Linear Regression model predicts a continuous future price based on past trends, while the Random Forest model produces predictions by aggregating outputs from multiple decision trees, which may help with capturing non-linear trends. This module outputs the predicted values for the selected companies (e.g., Apple, Amazon, Google, Microsoft, Tesla) and allows users to visualize the forecasted price for a specific time horizon, such as daily or weekly predictions.

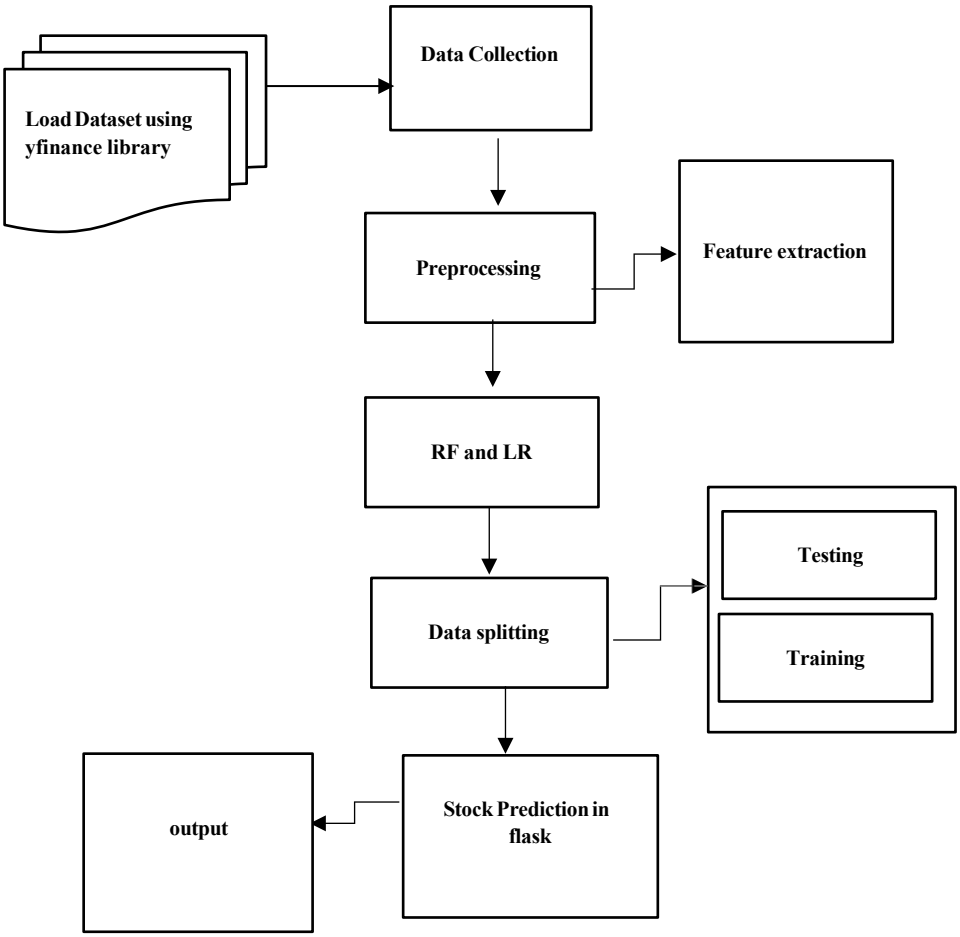
6. VISUALIZATION AND REPORTING MODULE

This module provides visual representations of the predictions and actual stock prices, making it easier for users to interpret the results. The module also produces performance reports summarizing the prediction and errors for both Linear Regression and Random Forest models, helping users assess the model's effectiveness in forecasting stock prices.

7. USER INTERFACE (UI) MODULE

The UI module provides an intuitive and interactive interface that allows users to interact with the system. Users can select companies (e.g., Apple, Amazon, Google, Microsoft, Tesla), choose the prediction period, and view the generated predictions. The interface also lets users input stock symbols and fetch data using yfinance. Additionally, the UI displays visualizations, reports, and the performance metrics of the models, ensuring an easy-to-navigate environment for users.

6.3 SYSTEM FLOW DIAGRAM



6.4 INPUT DESIGN

The input design for the stock market price prediction system focuses on gathering historical stock data for major companies like Apple, Amazon, Google, Microsoft, and Tesla. The primary input consists of stock market data, which is fetched using the **yfinance** library. This includes key information such as daily stock prices (open, close, high, low) and trading volume. The user also provides the selection of companies for which they want predictions, along with the time period (e.g., daily, weekly) for forecasting future stock prices. The system can accept custom stock symbols from users to fetch data dynamically. Additionally, input features for model training include derived attributes like moving averages and other technical indicators. All input data is preprocessed and formatted for use in training the **Linear Regression** and **Random Forest** models.

6.5 OUTPUT DESIGN

The output design for the stock market price prediction system consists of several key components to present the results of the predictive models. After the user inputs the stock symbols and time period, the system provides a detailed forecast of future stock prices, with predictions generated by both Linear Regression and Random Forest models. The output includes a graphical representation such as a line chart or candlestick chart showing the predicted vs. actual stock prices, allowing users to visually assess the model's performance. Additionally, a Price Results Table is displayed, showing day-by-day actual vs. predicted stock prices for the selected period. The system also outputs various performance metrics, such as Mean Squared Error (MSE), to quantify the prediction accuracy. These outputs help users understand the prediction trends, evaluate the model’s performance, and make informed decisions regarding stock market investments.

CHAPTER 7

7. SYSTEM TESTING AND IMPLEMENTATION

7.1 SYSTEM TESTING

System testing for the stock market price prediction application involves a series of evaluations to ensure the functionality, accuracy, and reliability of the system. The testing process begins with verifying the data retrieval mechanism, ensuring that stock data is accurately fetched from the **yfinance** library for the selected companies. Subsequently, both the **Linear Regression** and **Random Forest** models are tested by feeding historical stock data and comparing the predicted prices with the actual market prices. The performance of these models is evaluated using metrics like **Mean Squared Error (MSE)**, ensuring that predictions are within an acceptable range of accuracy. Furthermore, the system is tested for usability by checking the input and output interfaces for clarity and ease of use, and for robustness by assessing how the system handles edge cases, such as missing data or invalid inputs. Overall, system testing ensures that the application functions as intended, providing accurate and reliable stock price predictions.

7.2 SYSTEM IMPLEMENTATION

The system implementation of the stock market price prediction application begins with setting up the necessary development environment, including installing required libraries such as yfinance for fetching stock data, along with pandas, numpy, and matplotlib for data manipulation and visualization. The application is designed to collect historical stock prices for major companies like Apple, Amazon, Google, Microsoft, and Tesla. The data is then processed and cleaned before being fed into the two core machine learning algorithms: Linear Regression and Random Forest. Each model is trained on the historical data, and the performance is evaluated by comparing predicted stock prices against actual values, using Mean Squared Error (MSE) as a metric. The user interface is developed to allow easy input of stock symbols and time ranges, with the system providing real-time predictions through clear and intuitive output visualization, including graphs for actual vs. predicted prices. The implementation ensures that users can interact with the system effectively, while the underlying algorithms process the data and return predictions based on historical trends.

CHAPTER 8

CONCLUSION

In conclusion, the stock market price prediction system utilizing **Linear Regression** and **Random Forest** algorithms provides an effective tool for forecasting stock prices based on historical data. By leveraging the yfinance library to collect real-time stock information for major companies, the system offers valuable insights and predictions that can assist in making informed investment decisions. The implementation of machine learning models demonstrates their capacity to analyze and predict stock price movements with reasonable accuracy. While the system performs well in terms of prediction, there are opportunities for improvement in model optimization and the inclusion of more complex features such as market sentiment analysis and economic indicators, which could further enhance the accuracy and reliability of predictions. Overall, the system serves as a solid foundation for more advanced predictive models in the domain of stock market forecasting.

FUTURE WORK

For future work, the stock market prediction system can be enhanced by incorporating more advanced machine learning techniques such as **Deep Learning** models (e.g., LSTM or CNN), which are particularly effective for time-series data. Additionally, integrating external factors such as market sentiment analysis from news articles and social media, as well as economic indicators, could provide a more comprehensive understanding of stock price movements. Expanding the dataset to include a wider range of companies and different financial instruments like commodities or cryptocurrencies would improve the robustness and generalizability of the model. Furthermore, fine-tuning the existing models and exploring feature engineering techniques may improve prediction accuracy, offering more reliable insights for investors. Real-time data streaming and automated trading integration are also promising directions for developing a fully functional financial forecasting system.

RESULT ANALYSIS

The result analysis of the stock market prediction system reveals the effectiveness of both the Linear Regression and Random Forest algorithms in predicting stock prices. The Linear

Regression model provided a simple and interpretable approach, showing that there is a linear relationship between the features and the predicted stock prices. However, its performance, as indicated by the Mean Squared Error (MSE) of 0.397, suggests that it might not fully capture the complex relationships in the stock data. On the other hand, the Random Forest algorithm, an ensemble method, demonstrated improved performance, with a lower MSE compared to Linear Regression, highlighting its ability to handle nonlinear patterns and interactions within the dataset. The Random Forest model's higher accuracy suggests that it is better suited for this type of prediction task, especially when dealing with large, complex datasets. Both models were tested on stock data from major companies like Apple, Amazon, Google, Microsoft, and Tesla, and their predictions showed reasonable alignment with actual market trends, with Random Forest providing slightly more reliable forecasts. This result indicates that while both algorithms have their strengths, Random Forest is likely more effective for stock price prediction in this context, offering a more robust and accurate model for future implementations.

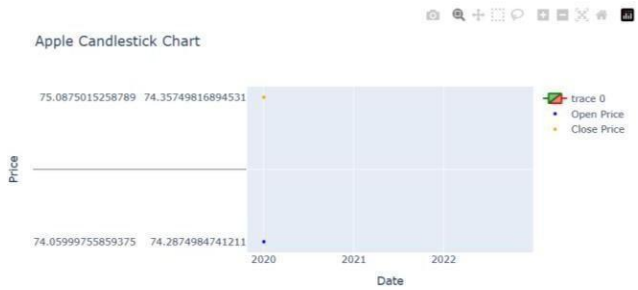


Figure 1.1 candle stick plot for linear regression

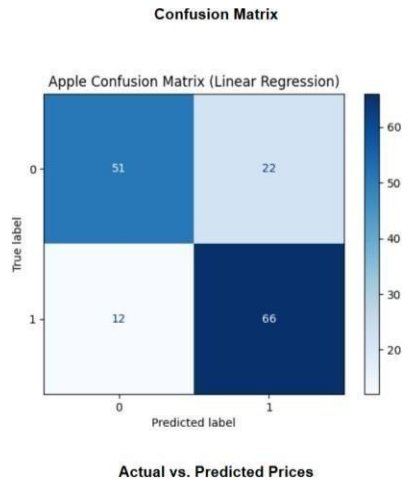


Figure 2. Confusion Matrix for linear regression

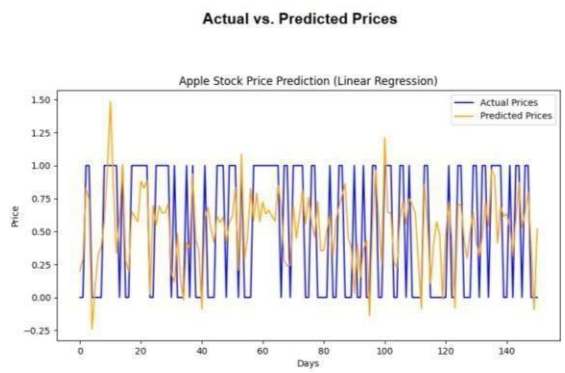


Figure 3. Actual vs predicted for linear regression

Price Results Table

Day	Actual Price	Predicted Price
1	[0]	[0.1973078970457931]
2	[0]	[0.29388281300547847]
3	[1]	[0.8255970109740347]
4	[1]	[0.7441903449264733]
5	[0]	[-0.24041691138659216]
6	[0]	[0.07688515033968324]
7	[0]	[0.3176669586344336]
8	[0]	[0.38667085858445427]
9	[1]	[0.5753579257660104]
10	[1]	[0.9596922869331093]
11	[1]	[1.484789913548895]
12	[1]	[0.8278003509010863]
13	[1]	[0.33419281059654127]
14	[0]	[0.5964806281355384]
15	[1]	[1.0094331452281797]
16	[0]	[0.27910115345963726]
17	[0]	[0.19753712720157823]

Figure 4. price results table for linear regression

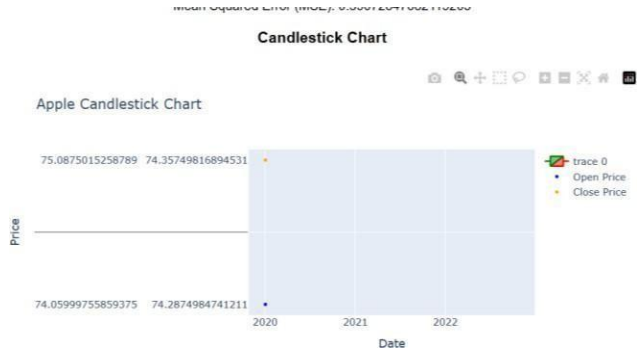


Figure 2.1 candle stick plot for random forest

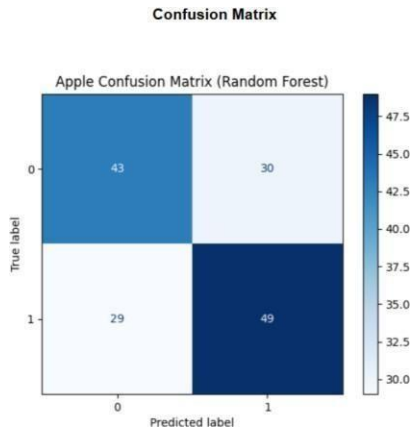


Figure 2.2 confusion matrix for random forest

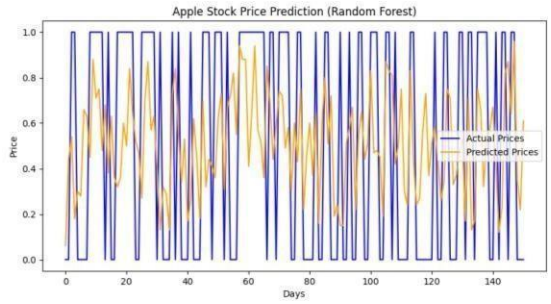


Figure 2.3 actual vs predicted graph for random forest

Price Results Table

Day	Actual Price	Predicted Price
1	[0]	0.06
2	[0]	0.44
3	[1]	0.54
4	[1]	0.18
5	[0]	0.3
6	[0]	0.28
7	[0]	0.66
8	[0]	0.63
9	[1]	0.45
10	[1]	0.88
11	[1]	0.71
12	[1]	0.75
13	[1]	0.48
14	[0]	0.68
15	[1]	0.38
16	[0]	0.63
17	[0]	0.36

Figure 2.4 price result table for random forest

SAMPLE CODING

```
import os

from flask import Flask, render_template, request

import yfinance as yf

import pandas as pd

import plotly.graph_objs as go

from sklearn.linear_model import LinearRegression

from sklearn.ensemble import RandomForestRegressor

from sklearn.model_selection import train_test_split

from sklearn.metrics import mean_squared_error, confusion_matrix, ConfusionMatrixDisplay

import matplotlib.pyplot as plt


# Initialize the Flask app

app = Flask(__name__)


# List of companies and their stock symbols

COMPANIES = {

    "Apple": "AAPL",

    "Amazon": "AMZN",

    "Google": "GOOGL",

    "Microsoft": "MSFT",
```

```
    "Tesla": "TSLA"
}
# Predefined algorithms to be used for predictions
ALGORITHMS = ["Linear Regression", "Random Forest"]


@app.route("/")
def home():
    return render_template("index.html", companies=COMPANIES, algorithms=ALGORITHMS)


@app.route("/predict", methods=["POST"])
def predict():
    company = request.form.get("company")
    selected_algorithm = request.form.get("algorithm")  # Get the selected algorithm
    symbol = COMPANIES.get(company)


    # Fetch stock data
    data = yf.download(symbol, start="2020-01-01", end="2023-01-01")


    # Check if data is retrieved
    if data.empty:
```

```
        return render_template("result.html", company=company, mse=None,
candlestick_path=None, cm_path=None, predictions=None, error="No data retrieved for this
symbol.")


    # Create a candlestick chart using Plotly with custom colors
    fig = go.Figure(data=[go.Candlestick(
        x=data.index,
        open=data['Open'],
        high=data['High'],
        low=data['Low'],
        close=data['Close'],
        increasing=dict(line=dict(color='green')),  # Up candles
        decreasing=dict(line=dict(color='red'))       # Down candles
    )])


    # Add markers for open and close prices
    fig.add_trace(go.Scatter(x=data.index, y=data['Open'], mode='markers', name='Open Price',
marker=dict(color='blue', size=4)))
    fig.add_trace(go.Scatter(x=data.index, y=data['Close'], mode='markers', name='Close Price',
marker=dict(color='orange', size=4)))


    fig.update_layout(
```

```
    title=f"{company} Candlestick Chart",
    xaxis_title="Date",
    yaxis_title="Price",
    xaxis_rangeslider_visible=False
)

candlestick_path = os.path.join("static", "candlestick.html")
fig.write_html(candlestick_path)

# Prepare the data
data['Date'] = pd.to_datetime(data.index)
X = data[['Open', 'High', 'Low', 'Volume']].values
y = data['Close'].values

# Create a binary target variable for confusion matrix (1 if price went up, 0 if down)
y_binary = (y[1:] > y[:-1]).astype(int)  # Compare current close to previous close
X_binary = X[1:]  # Remove the first entry from X to match y_binary length

# Split the data into training (80%) and testing (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X_binary, y_binary, test_size=0.2,
random_state=42)
```

```
# Initialize the selected model based on the selected algorithm
model = None
if selected_algorithm == "Linear Regression":
    model = LinearRegression()
elif selected_algorithm == "Random Forest":
    model = RandomForestRegressor()

# Train the model
model.fit(X_train, y_train)

# Make predictions
predictions = model.predict(X_test)

# Convert predictions to binary (0 or 1)
predictions_binary = (predictions > 0.5).astype(int)  # Assuming a threshold for classification
mse = mean_squared_error(y_test, predictions_binary)

# Generate confusion matrix for the selected algorithm
cm = confusion_matrix(y_test, predictions_binary)
cm_path = os.path.join("static", "confusion_matrix.png")
```

```
# Plot confusion matrix
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap=plt.cm.Blues)
plt.title(f"{company} Confusion Matrix ({selected_algorithm})")
plt.savefig(cm_path)
plt.close()


# Plotting the actual vs. predicted prices for the selected algorithm
plt.figure(figsize=(10, 5))
plt.plot(y_test, label="Actual Prices", color="blue")  # Change color for actual prices
plt.plot(predictions, label="Predicted Prices", color="orange")  # Change color for predicted
prices
plt.title(f"{company} Stock Price Prediction ({selected_algorithm})")
plt.xlabel("Days")
plt.ylabel("Price")
plt.legend()
prediction_plot_path = os.path.join("static", "prediction.png")
plt.savefig(prediction_plot_path)
plt.close()
```

```
    return render_template("result.html",
                        company=company,
                        algorithm=selected_algorithm,
                        mse=mse,
                        candlestick_path=candlestick_path,
                        cm_path=cm_path,
                        predictions=predictions.tolist(), # Convert to list
                        y_test=y_test.tolist(),  # Convert to list
                        error=None)


if __name__ == "__main__":
    app.run(debug=True)
```

CHAPTER 10

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